

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\mu_1(2\varepsilon - \varepsilon^2)\alpha_1^2 &= \frac{1-c}{t}(2\varepsilon - \varepsilon^2)\frac{1}{2t}\left(\sum_{v\in V(K)}\delta_v\right)^2 \\ &< \varepsilon t^{-2}\left(\sum_{v\in V(K)}\delta_v\right)^2 \\ &\leq \min\left\{\varepsilon^{-1}t^{-2}\left(\sum_{v\in S}\delta_v\right)^2, 2\varepsilon t^{-1}\sum_{v\in V(K)}\delta_v^2\right\}\end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{array}{ccccccc} & & L_{-1} & L_0 & L_1 & L_2 & \ldots \\ & J_{-2} & J_{-1} & J_0 & J_1 & J_2 & \ldots \\ \Lambda_{-3} & \Lambda_{-2} & \Lambda_{-1} & \Lambda_0 & \Lambda_1 & \Lambda_2 & \ldots \\ \text{.....} & & & \ldots & & \text{.....} & \end{array}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}&\int\limits_0^t\int\limits_{\mathbb{R}^3}\partial_t\varphi_1\xi+\gamma k_1v\cdot(\xi\nabla\varphi_1+\varphi_1\nabla\xi)+\gamma k_1\bar{p}v\cdot\nabla\varphi_1-k_1\varphi_1v\cdot\nabla\xi\,\mathrm{d}x\mathrm{d}s\\&beginalign*6pt]=-\int\limits_{\mathbb{R}^3}\xi_0(x,\tau)\varphi_1(\cdot,0)\,\mathrm{d}x.\end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\delta k &= \tfrac{1}{2}\theta k\ell, & \delta A^{(1)} &= -\theta A^{(2)}, \\ \delta e^\phi &= -\tfrac{7}{4}\theta\ell e^\phi, & \delta A^{(2)} &= \theta A^{(1)}, \\ \delta\ell &= \theta\left(1+\ell^2-2ke^{-2\phi}\right), & \delta B &= 0,\end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
P_0 U(\vec{a}, 1) B_s^n 1 &= P_0 \psi^{\beta_1, n}(g_1)_{U(\vec{a})} \psi^{\beta_2, n}(g_2)_{U(\vec{a})} \cdots \psi^{\beta_{s-1}, n}(g_{s-1})_{U(\vec{a})} U(\vec{a}, 1) \psi^{\beta_s, n}(g_s) 1 \\
&= e^{i[a^0 \hat{H} - a^1 \hat{P}]} P_0 B_s^{n, \vec{a}} 1 = e^{i[a^0 \hat{H} - a^1 \hat{P}]} \left[B_s^{n, \vec{a}} 1 - \langle B_s^n 1, 1 \rangle 1 \right],
\end{aligned}$$